Pinned production functional: unique zero equilibrium and global minimizer

TECT verification-first repository
claim `A2-FULL-PRODUCTION-WELLPOSED`

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1. Purpose and scope: a decisive candidate test

The current production functional is one candidate microscopic law, not an axiom and not an empirically selected vacuum theory. Nevertheless its own global minimizer can be decided exactly at the pinned parameters. On the fixed three-torus and under the named A2 hypothesis identifying the hash-pinned P1 functional as the canonical continuum functional, this note proves

$$\boxed{\mathcal{F}_{\mathrm{P1}}[\Psi] \geq g\,\|\Psi\|_{L^2}^2, \qquad g = \frac{719818750025582338837}{5400000000000000000000} > \frac{1}{8}.} \tag{1.1}$$

In addition,

$$\langle D\mathcal{F}_{\mathrm{P1}}(\Psi), \Psi\rangle \geq \kappa \|\Psi\|_{L^2}^2, \qquad \kappa = \frac{2101675000076747016511}{8100000000000000000000} > \frac{1}{4}. \tag{1.2}$$

Since $\mathcal{F}_{\mathrm{P1}}[0] = 0$, the zero field is the unique global minimizer; (1.2) also makes it the unique critical point. For the canonical A2 gradient flow $\partial_t \Psi = -D\mathcal{F}(\Psi)$ it further gives the exponential estimate

$$\|\Psi(t)\|_{L^2}^2 \leq e^{-2\kappa t}\|\Psi(0)\|_{L^2}^2. \tag{1.3}$$

Thus no BCC, lamellar, other finite-wave-number pattern, uniform nonzero condensate, or unconstrained nonzero field can be the global ground state of this exact pinned functional. A numerical phase tournament cannot reverse this inequality.

This is a theorem about the declared classical P1/A2 functional. It is not a physical-vacuum theorem and does not transfer to the historical nonvariational backend, the signed A7 covariance-normal stochastic composite, another parameter branch, or another candidate equation. It does not exclude transients outside the canonical gradient flow, fixed-charge or fixed-norm constraints, compact targets, chemical-potential modifications, conserved dynamics, or other nonequilibrium equations.

2. Exact quadratic lower bound

Write $\rho = |\Psi|^2$. The scalar Fourier symbol in each internal direction is

$$D(x) = Yx^2 + Zx + r, \qquad x = |k|^2 \geq 0.$$

The pinned values have $Y = 1$ and $Z < 0$, so completion of the square gives

$$D(x) = Y\left(x + \frac{Z}{2Y}\right)^2 + \mu_{\mathrm{sh}}, \qquad \mu_{\mathrm{sh}} = r - \frac{Z^2}{4Y} = \frac{26000000000947494031}{10^{20}} > 0. \tag{2.1}$$

This is the P1 production shell minimum. It must not be confused with the different scalar-slice manifest value used by the historical N-001 kernel audit.

The family and lock matrix is

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}.$$

For $M - (7/250)\mathrm{I}$, the three leading Sylvester minors are

$$\frac{9}{125}, \qquad \frac{1211}{250000}, \qquad \frac{89}{31250000}. \tag{2.2}$$

all strictly positive. Hence $M > (7/250)\mathrm{I}$. Parseval therefore yields

$$\mathcal{F}_{\text{quad}}[\Psi] \geq \frac{\nu}{2}\|\Psi\|_{L^2}^2, \qquad \nu = \mu_{\text{sh}} + \frac{7}{250} = \frac{28800000000947494031}{10^{20}}. \tag{2.3}$$

3. Class-II positivity and local completion

The Class-II current pair has coefficient matrix

$$Q_{\text{II}} = \frac{1}{M_X^2 + \epsilon_X}\begin{pmatrix} c_{JJ}\alpha_X^2 & c_{JK}\alpha_X\beta_X \\ c_{JK}\alpha_X\beta_X & c_{KK}\beta_X^2 \end{pmatrix}.$$

Write $P = M_X^2 + \epsilon_X > 0$. Its leading entry is positive and

$$c_{JJ}c_{KK} - c_{JK}^2 = \frac{1}{50} > 0. \tag{3.1}$$

Therefore $Q_{\text{II}} > 0$ and the full Class-II energy is nonnegative. The registered shell-bias coefficient is exactly $\eta_{\text{shell}} = 0$.

It remains only to combine the negative quartic with the positive sextic. For the pinned $\lambda = -43/100$ and $\gamma = 81/50$,

$$\begin{aligned} \frac{\nu}{2} + \frac{\lambda}{4}\rho + \frac{\gamma}{6}\rho^2 &= g + \frac{\gamma}{6}(\rho - \rho_*)^2, \\ \rho_* &= -\frac{3\lambda}{4\gamma} = \frac{43}{216}, \\ g &= \frac{\nu}{2} - \frac{3\lambda^2}{32\gamma} = \frac{719818750025582338837}{5400000000000000000000} > \frac{1}{8}. \end{aligned} \tag{3.2}$$

Multiplication by ρ, spatial integration, and addition of the nonnegative Class-II and unused quadratic-square terms prove (1.1). The strict coefficient $g > 0$ also makes equality possible only for $\rho = 0$ almost everywhere.

There is a second exact sign which rules out hidden nonzero equilibria. Along the amplitude ray $\Psi \mapsto t\Psi$, put $y = t^2$. At a point write the unscaled Class-II currents as j and $\ell(y) = j - q(y)\nabla\rho$, where

$$q(y) = \frac{ym}{y\rho + \epsilon}, \qquad \theta = \frac{\epsilon}{y\rho + \epsilon} \in [0, 1].$$

Direct differentiation gives

$$\frac{dE_{\text{II}}}{dy} = y\begin{pmatrix} j & \ell \end{pmatrix} R_\theta \begin{pmatrix} j \\ \ell \end{pmatrix}, \quad R_\theta = \begin{pmatrix} a - \theta b & b + \theta(b-c)/2 \\ b + \theta(b-c)/2 & c(1+\theta) \end{pmatrix}. \tag{3.3}$$

Here the displayed two-current form is applied componentwise, with $|j|^2$, $\mathrm{Re}(\bar{j}\ell)$ and $|\ell|^2$, and is summed over the internal generators and spatial directions. For the pinned coefficients, $a - \theta b \geq 21/(2000P) > 0$ and

$$\det R_\theta = \frac{9(-81\theta^2 + 128\theta + 128)}{10240000P^2} > 0 \quad (0 \leq \theta \leq 1), \tag{3.4}$$

because the numerator is concave and positive at both endpoints. Hence the Class-II energy is nondecreasing on every amplitude ray. The remaining radial derivative satisfies

$$\langle D\mathcal{F}(\Psi), \Psi \rangle \geq \left(\nu - \frac{\lambda^2}{4\gamma}\right) \|\Psi\|_{L^2}^2 = \kappa \|\Psi\|_{L^2}^2, \tag{3.5}$$

which proves (1.2). Combining (1.2) with $\frac{1}{2}\frac{d}{dt}\|\Psi\|_{L^2}^2 = -\langle D\mathcal{F}(\Psi), \Psi \rangle$ proves (1.3).

4. Consequences for the phase-neutral model tournament

The conclusion is stronger than a finite BCC-versus-flat comparison:

$$\mathcal{F}_{\mathrm{P1}}[\Psi] - \mathcal{F}_{\mathrm{P1}}[0] > 0 \quad \text{for every nonzero } \Psi \in H^2(\mathbb{T}_L^3; \mathbb{C}^3). \tag{4.1}$$

Accordingly the pinned M1 functional cannot realize the founding requirement of a stable ordered state lying below its declared zero reference state. This is evidence for rejecting or retuning this implementation, not evidence that some preferred lattice should replace it. M0 and structurally different candidate families remain logically untouched.

The radial derivative theorem also rules out every nonzero stationary point and hence every nonzero equilibrium local minimum. T-052's finite-star, multistart, Hessian, and metastability search is therefore superseded for this exact M1 functional. It may be retained only as a backend regression test; it cannot discover a missing equilibrium branch without falsifying R-157.

5. Devil's-advocate audit

1. **Objection: the negative gradient coefficient invalidates the quadratic lower bound. DISMISSED.** Equation (2.1) is the exact continuous square completion and the torus Fourier set is a subset of $x \geq 0$.

2. **Objection: the Class-II cross term may be negative. DISMISSED in the pinned P1 functional.** The full two-by-two coefficient matrix is positive definite by (3.1); no current term is dropped.

3. **Objection: Young coercivity proves only boundedness below, not a unique minimizer. DISMISSED.** The sharper exact completion (3.2) leaves the strictly positive multiple $g\rho$, so every nonzero field has strictly positive energy.

4. **Objection: this proves the physical vacuum is empty. UPHELD against that inference.** The theorem decides only a candidate functional whose physical selection remains unproved. It is a reason to reject or revise that candidate, not to infer that Nature has no condensate.

5. **Objection: the signed A7 stochastic composite is also positive. UPHELD as false.** Covariance-normal subtraction makes the A7 owner signed. R-157 does not modify A13/T-050 and cannot be used as its proof.

6. **Objection: the energy lower bound alone cannot remove nonzero saddles or local minima. DISMISSED by a separate argument.** Equations (3.3)–(3.5), not the energy bound alone, give a strictly positive radial derivative at every nonzero field and therefore exclude all nonzero critical points.

7. **Objection: radial scaling is illegal in every plausible TECT order-parameter space. UPHELD as a scope boundary, not against this theorem.** The A2 domain is the unconstrained linear space $H^2(\mathbb{T}^3_L;\mathbb{C}^3)$, where radial scaling is admissible. A fixed norm or charge, a chemical-potential Legendre transform, a compact $\mathbb{CP}^2$ target, or conserved dynamics changes the variational problem and remains an alternative candidate rather than a consequence of R-157.

External review is invited for the real-pairing factor 1/2, the Parseval normalization, the distinction between the P1 shell minimum and the historical scalar-slice mass, the Sylvester lower bound, the Class-II determinant, the quartic-sextic completion, and every scope firewall above.

6. Result footer and reproduction

Run from the repository root:

```
$py = "E:\Dev\TECT.venv\Scripts\python.exe"
& $py codes/foundations/a2_pinned_functional_unique_zero_global_minimizer_verify.py
```

The verification package has three deliberately different layers:

Layer	Method	Required result
Primary	exact SymPy derivation plus backend fixtures	26/26 PASS
Independent	standard-library rational reconstruction	24/24 PASS
Integrated	authority, artifact, PDF, and release audit	all PASS

The existing A2 well-posedness theorem remains the continuum-domain authority. R-157 extends its pinned-functional conclusion without changing its tier: the same named A1 identification hypothesis remains in force. The result changes the candidate verdict, not the empirical status of the model.

```
Result ID: A2-PINNED-FUNCTIONAL-UNIQUE-ZERO-GLOBAL-MINIMIZER
Ledger ID: R-157
Claim: A2-FULL-PRODUCTION-WELLPOSED.
Precise statement: F_P1[Psi] >= g ||Psi||_L2^2 with exact g>1/8 and
  <DF_P1(Psi),Psi> >= kappa ||Psi||_L2^2 with exact kappa>1/4;
  hence Psi=0 is the unique critical point and global minimizer, and the
  canonical L2 gradient flow decays at rate 2 kappa.
Scope: Unconstrained hash-pinned P1/A2 continuum functional,
  eta_shell=0, H2 fields, and its canonical gradient flow.
Dependencies: A1-PRODUCTION-FUNCTIONAL-REALISATION, the named
  A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL hypothesis, and the A2 continuum
  theorem.
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED, CONDITIONAL.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a2_pinned_functional_unique_zero_global_minimizer_verify.py
Expected output: Integrated verifier PASS; primary 26/26; independent 24/24;
  legacy A2 61/61.
Falsification gate: Any exact shell/internal/potential/Class-II sign, radial
  derivative, child parity, authority hash, record, PDF, or legacy A2 gate
  fails, or a nonzero pinned critical point is produced.
Tier before / after: T6 / T6; no promotion.
No-overclaim statement: No physical-vacuum, historical-backend, signed-A7,
  constrained/compact-target, conserved/chemical-potential,
  general-nonequilibrium, alternative-parameter, alternative-model,
  T-050/A13, or Sector-A conclusion follows.
Next required action: Use R-157 as the M1 early-rejection boundary in T-054;
  T-052 is superseded except as regression. Keep A13/T-050 separate.
```